

Explaining the Paper Simply

Why Japan's incineration energy-recovery pattern matters

8-10 MINUTE LIVE ROUTE

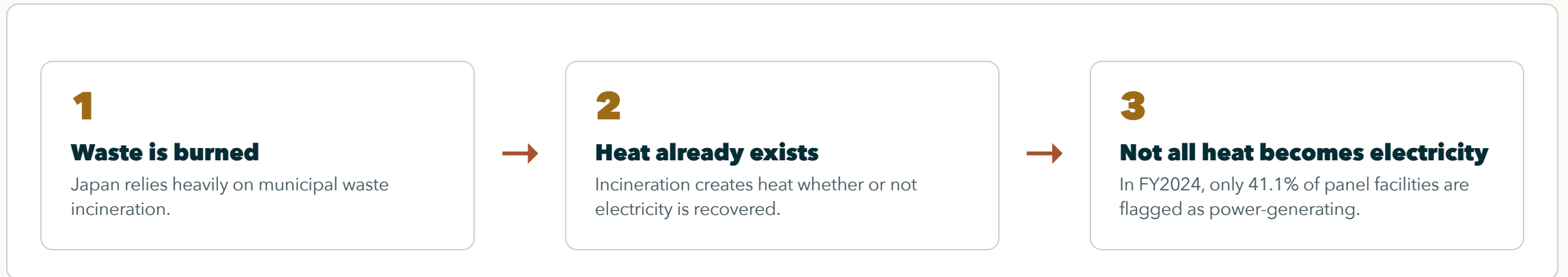
PAPER ONLY

APPENDIX ONLY IF ASKED

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Paper briefing for discussion and feedback

Why This Matters



This matters because the same waste-treatment process can either recover useful power or miss that energy-recovery opportunity.

Main Idea in Plain Words

PLAIN CLAIM

One fleet average hides two different questions.

QUESTION 1

Which plants start generating electricity?

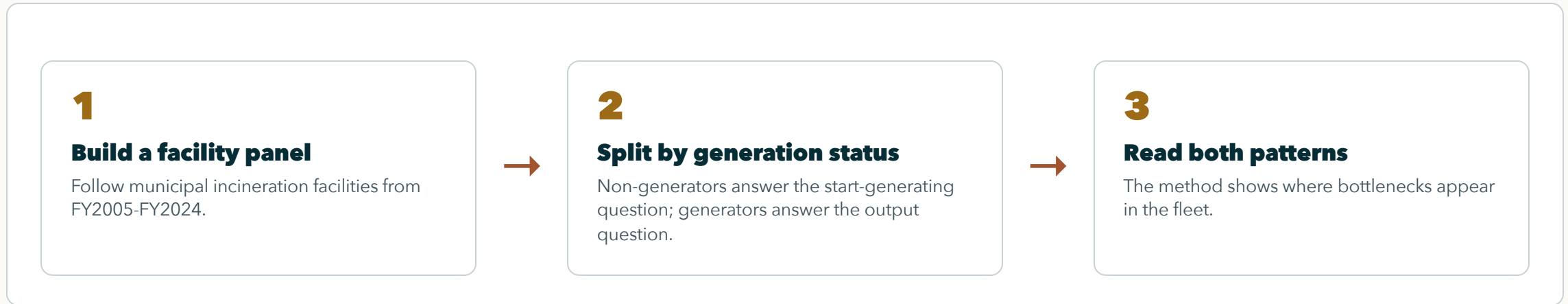
This is the "entry" question.

QUESTION 2

Among generators, who produces more electricity per tonne?

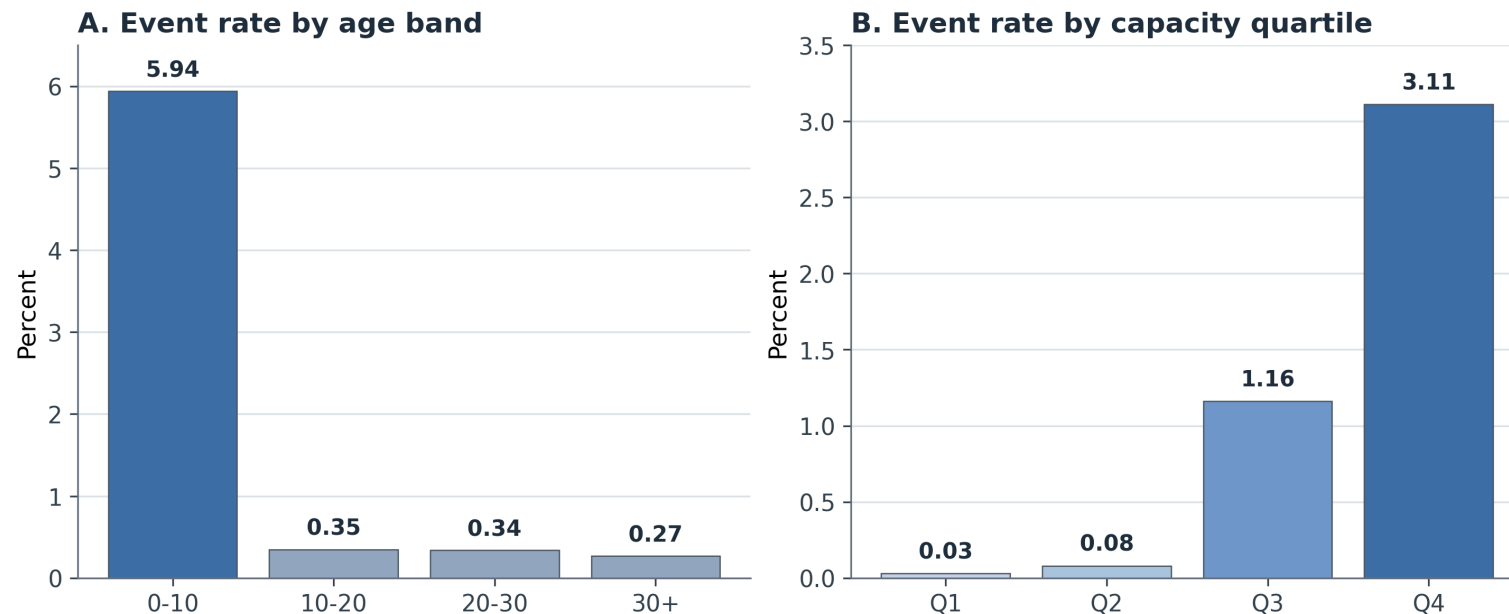
This is the "performance" question.

Method in Plain Language



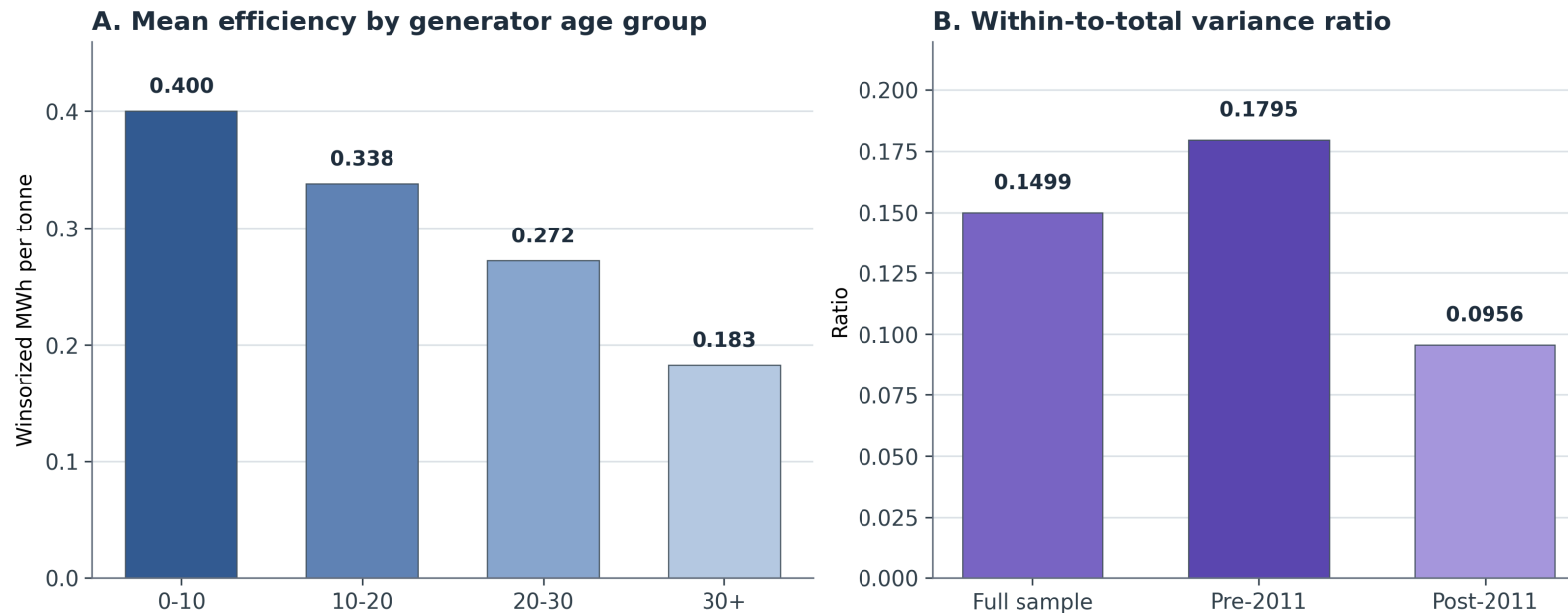
This is diagnostic: it identifies patterns and bottlenecks, not strict cause-and-effect.

Result 1: Starting Generation Is Uneven



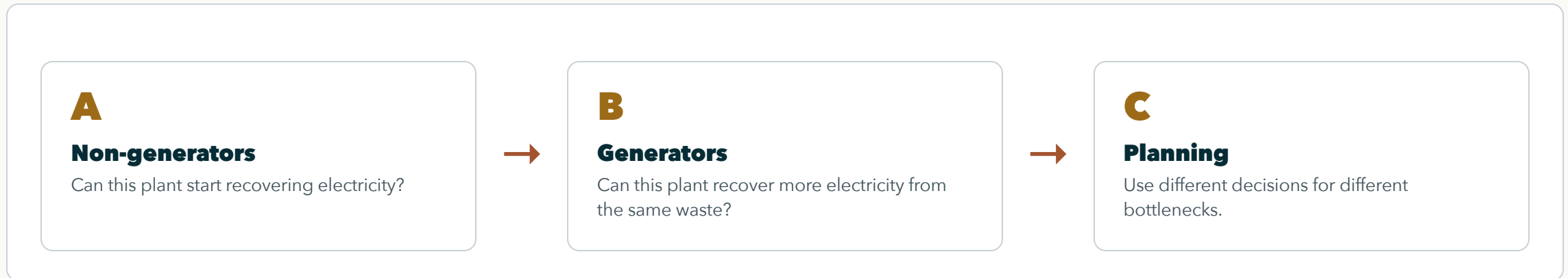
Facilities that first start generating electricity are mostly young and large: 102/141 events are from age 0-10 facilities, and 99/141 are in the largest capacity quartile.

Result 2: Generator Output Is Uneven



Among plants that already generate, older facilities recover less electricity per tonne. Scale and utilization help, but they do not erase large facility-level differences.

Why the Method Matters for Decisions



This matters because renewal, starting electricity generation, and generator optimization are not the same task.

Why This Can Stand as a Paper

CLEAR QUESTION

It separates starting generation from generator output.

That avoids a confusing one-average story.

USEFUL EVIDENCE

It follows one national facility panel.

The same fleet is read through two linked questions.

CAREFUL CLAIM

It says what the evidence can and cannot prove.

That keeps the paper useful without overstating causality.

What This Does and Does Not Prove

BOUNDARIES

- No strict causal policy-effect claim.
- No proof that replacement is the only pathway.
- No complete lifecycle climate accounting.
- No claim that the generator frame is a perfect census.

DEFENDED CLAIM

It identifies where the bottlenecks appear.

Starting generation is unevenly spread, and generator output remains uneven after entry.

Feedback Ask

The paper's message is: first ask who starts generating electricity, then ask who generates well after starting.

Useful feedback: is this two-part story clear, defensible, and worth using as the paper's main pitch?

Appendix A: Method Details

START-GENERATING MODEL

Among facilities not yet generating, estimate who first reports generation in the next observed year.

- Outcome: first report of power generation in the next observed year.
- Predictors: prior-year age band and design capacity.
- Controls: fiscal-year fixed effects and prefecture fixed effects.
- Uncertainty: facility-clustered standard errors.

GENERATOR-OUTPUT MODEL

Among identifiable operating generators, compare electricity generated per tonne processed.

- Outcome: winsorized log electricity generated per tonne processed.
- Predictors: age, capacity, utilization, heating value, grid-emission control.
- Models: pooled OLS, year FE, RE, and year FE plus RE.
- Uncertainty: facility-clustered standard errors.

Appendix B: Key Numbers

Evidence item	Number	Interpretation
FY2024 facilities flagged as power-generating	41.1%	Most facilities remain outside electricity generation in the panel.
Source panel	23,599	Facility-year observations across FY2005-FY2024.
Identifiable facilities	2,948	National facility-level evidence.
Adoption risk-set size	13,770	Facility-years first observed without generation.
Observed first-adoption events	141	Events used to describe entry into generation.
Events from age 0-10 facilities	102	Adoption is concentrated among young facilities.
Events in largest capacity quartile	99	Adoption is concentrated among large facilities.
Generator regression frame	5,683	Identifiable operating-generator observations.
Within-to-total variance ratio	0.1499	Most generator-efficiency variation is between facilities.

Appendix C: Pathway Audit

RESET / REBUILD-LIKE

82

observed events

CONTINUITY / UPGRADE-LIKE

38

observed events

PLACEHOLDER / FORWARD-DATED

20

observed events

Safe wording: selective modernization with capital-intensive evidence, not proof that replacement is the only pathway.

Appendix D: File Map

SLIDES

`paper/slides/paper-zoom-briefing.md`

Editable source deck.

SCRIPT

`paper/slides/paper-zoom-script.md`

Separate live presentation script.

CHECKLIST

`paper/slides/paper-zoom-presentation-checklist.md`

Run sheet for a simple, direct presentation.

BUILD COMMAND

`npm run slides:paper:pdf`

Regenerates the shareable PDF.